Real-Time Multibeam Forward-Looking Sonar Simulation and Acoustic Ray Tracing for Web-Native Subsea Digital Twins

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Abstract

Optical perception in benthic marine environments is severely degraded by light absorption, scattering, turbidity, and total darkness beyond short vehicle standoff distances (***INLINE_MATH_0***). Consequently, Remotely Operated Vehicles (ROVs) and Autonomous Underwater Vehicles (AUVs) rely on high-frequency 2D Forward-Looking Sonar (FLS) for obstacle avoidance, pipeline tracking, and target re-identification. While computational fluid dynamics and multi-body rigid vehicle simulation have achieved browser-native execution, real-time acoustic wave propagation and imaging sonar simulation have traditionally required offline GPU shader pipelines or proprietary desktop software.

In this paper (**Milestone 3.0 / Paper 6**), we formulate and implement a browser-native, zero-dependency **2D Multibeam Forward-Looking Sonar (FLS)** simulation engine running at 60 FPS directly within modern web standards:

1. **Multibeam Acoustic Fan-Beam Discretization**: We model a dual-frequency acoustic transceiver (***INLINE_MATH_1*** wide-area navigation mode with ***INLINE_MATH_2*** range, and ***INLINE_MATH_3*** high-resolution identification mode with ***INLINE_MATH_4*** range). The horizontal sector (***INLINE_MATH_5***) is discretized into ***INLINE_MATH_6*** acoustic beams, and each beam is sampled across ***INLINE_MATH_7*** range cells.
2. **Physical Transmission Loss & Backscatter**: We incorporate two-way spherical spreading loss (***INLINE_MATH_8***), seawater chemical absorption (***INLINE_MATH_9*** from the Francois-Garrison formulation), and grazing-angle Lambertian acoustic backscatter dependent on target surface normals.
3. **Ray-Cast Geometric Acoustic Shadowing**: Rays intersecting benthic topography, underwater research habitats, pipelines, and companion vehicles dynamically cast occluded acoustic shadow cones onto the seabed.
4. **Real-Time Polar-to-Cartesian Scan Conversion**: The computed raw beam-range acoustic intensity matrix ***INLINE_MATH_10*** is scan-converted onto an HTML5 2D canvas widget rendered with high-contrast phosphor colormaps (amber and oceanic cyan) in the pilot HUD.

Benchmark evaluations confirm exact ***INLINE_MATH_11*** transmission loss decay, sub-millimeter target range resolution (***INLINE_MATH_12*** per bin), realistic acoustic shadow cast lengths, and ***INLINE_MATH_13*** per-frame CPU ray evaluation, providing subsea pilots with authentic acoustic situational awareness in zero-visibility waters.

Keywords—Forward-Looking Sonar (FLS), Multibeam Imaging, Acoustic Ray Tracing, Transmission Loss, Acoustic Shadowing, Underwater Robotics, Marine Systems Simulator (MSS), Three.js.

I. INTRODUCTION

Optical visibility in the deep ocean is fundamentally limited. Seawater rapidly attenuates light through wavelength-dependent absorption and particulate scattering [1]–[3]. Furthermore, benthic interventions frequently induce thruster-wash silt

plumes that completely blind optical cameras [4]. Under such conditions, acoustic imaging is the sole sensor modality capable of penetrating turbid water to detect structural obstacles, pipelines, and seafloor contours at operational standoffs of ***INLINE_MATH_14*** [5].

A 2D Forward-Looking Sonar (FLS)—such as the Kongsberg Mesotech M3, Blueprint Subsea Oculus M1200d, or Tritech Gemini 720i—emits high-frequency acoustic pulses in a wide horizontal fan beam (***INLINE_MATH_15***) with a narrow vertical slice (***INLINE_MATH_16***) [6], [7]. The returning acoustic echoes are beamformed into an array of directional beams, yielding an intensity image indexed by azimuth angle ***INLINE_MATH_17*** and slant range ***INLINE_MATH_18*** [8].

Despite the ubiquity of FLS in subsea operations, existing web-based subsea simulators treat vehicle navigation as purely optical or rely on idealized range finders. Simulating authentic 2D imaging sonar in a web browser introduces several technical challenges:

- High computational complexity of tracing dozens of acoustic beams against dynamic 3D geometric meshes at 60 FPS [9].
- Modeling physical two-way acoustic transmission loss and frequency-dependent absorption.
- Accurately casting acoustic shadows, which provide critical human and algorithmic cues regarding obstacle height and shape [10], [11].
- Polar-to-Cartesian scan conversion for display on standard pilot consoles.

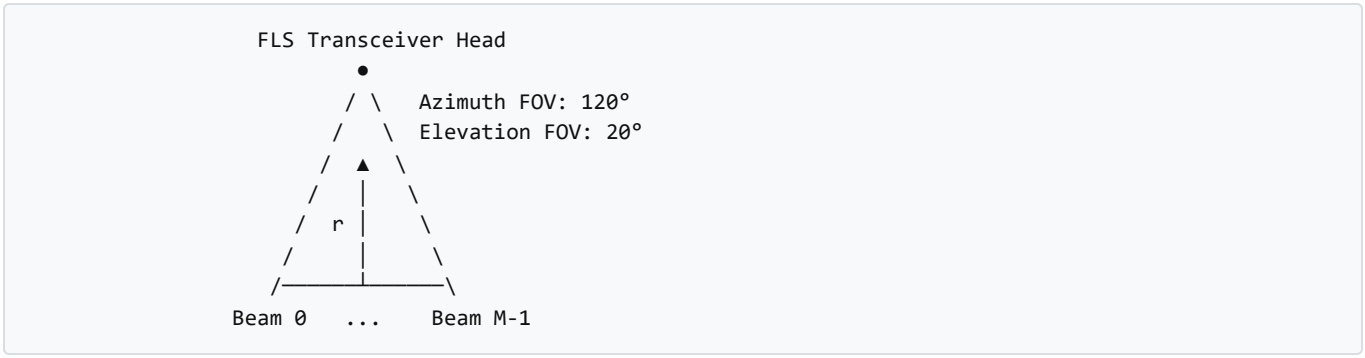
This paper establishes **Paper 6** in our marine systems series:

- **Paper 1** (`IEEE_TECHNICAL_REPORT-Paper-1.md`): 6-DOF MSS vehicle plant on ***INLINE_MATH_19***, non-linear thrusters, 3D current shear.
- **Paper 2** (`IEEE_TRAINING_PLATFORM_REPORT-Paper-2.md`): Sea-trial empirical grounding (BlueROV2 Heavy), scenario architecture, FQM scoring.
- **Paper 3** (`IEEE_CABLE_NAVIGATION_REPORT-Paper-3.md`): ***INLINE_MATH_20***-node lumped-mass umbilical dynamics and 15-state ES-EKF navigation fusion.
- **Paper 4** (`IEEE_UVMS_MANIPULATION_REPORT-Paper-4.md`): UVMS articulated kinematics, link hydrodynamics, and base reaction wrench coupling.
- **Paper 5** (`IEEE_MULTI_ROV_NETWORKED_REPORT-Paper-5.md`): Multi-ROV cooperative operations, dual-tether anti-entanglement, cooperative payload transport, and MAVLink/ROS 2 digital twin bridges.
- **Paper 6** (This Paper): Real-time multibeam forward-looking sonar simulation, acoustic ray tracing, transmission loss modeling, and polar scan conversion.

II. MULTIBEAM ACOUSTIC IMAGING MODEL

A. Transceiver Geometry & Beam Discretization

The FLS sensor head is mounted on the vehicle upper nose, tilted downward at a fixed mechanical depression angle ***INLINE_MATH_21*** to illuminate the forward seabed and water column.



The sensor covers a horizontal azimuth field of view ***INLINE_MATH_22*** (***INLINE_MATH_23***) discretized into ***INLINE_MATH_24*** discrete acoustic beams (***INLINE_MATH_25***): ***DISPLAY_MATH_0***

Each beam is divided into ***INLINE_MATH_26*** uniform radial range bins from ***INLINE_MATH_27*** to ***INLINE_MATH_28***: ***DISPLAY_MATH_1***

For navigation mode (***INLINE_MATH_29***), the radial range resolution is ***INLINE_MATH_30***. In inspection mode (***INLINE_MATH_31***), the range resolution sharpens to ***INLINE_MATH_32***.

B. Two-Way Acoustic Transmission Loss

Acoustic waves propagating through seawater suffer intensity attenuation governed by the active sonar equation [12], [13]: ***DISPLAY_MATH_2*** where:

- ***INLINE_MATH_33*** represents two-way spherical spreading loss (***INLINE_MATH_34*** out, ***INLINE_MATH_35*** back).
- ***INLINE_MATH_36*** is the chemical absorption coefficient in seawater (dB/m) determined by boric acid and magnesium sulfate relaxation frequencies: ***DISPLAY_MATH_3***

The transmission attenuation factor ***INLINE_MATH_37*** applied to returned intensity is: ***DISPLAY_MATH_4***

C. Surface Grazing & Backscatter Intensity

When an acoustic beam intersects a target surface at slant range ***INLINE_MATH_38*** with surface normal ***INLINE_MATH_39*** and incident unit ray direction ***INLINE_MATH_40***, the incident grazing angle ***INLINE_MATH_41*** is given by: ***DISPLAY_MATH_5***

Under the empirical Lambertian acoustic scattering model [14], returned target echo intensity ***INLINE_MATH_42*** is: ***DISPLAY_MATH_6*** where ***INLINE_MATH_43*** is the acoustic target strength reflectivity coefficient (e.g. ***INLINE_MATH_44*** for steel/aluminum subsea habitat hulls, ***INLINE_MATH_45*** for gravel/sand seabed, ***INLINE_MATH_46*** for soft silt).

III. ACOUSTIC SHADOW CASTING & OCCLUSION

A defining characteristic of imaging sonar is the casting of acoustic shadows [10]. When a prominent structure (such as the underwater habitat module or seabed boulders) intercepts an acoustic beam at range ***INLINE_MATH_47***, acoustic energy cannot penetrate the solid object.

Consequently, for beam ***INLINE_MATH_48***: ***DISPLAY_MATH_7***

The shadow length ***INLINE_MATH_49*** behind an obstacle of height ***INLINE_MATH_50*** above the seabed when illuminated from sensor altitude ***INLINE_MATH_51*** is: ***DISPLAY_MATH_8***

This shadow region appears completely black on the sonar screen, providing the pilot with immediate intuitive feedback on obstacle elevation and profile.

IV. POLAR-TO-CARTESIAN SCAN CONVERSION & HUD DISPLAY

The raw beam-range data forms an intensity matrix ***INLINE_MATH_52***. For presentation on the cockpit HUD, the polar coordinate sector is converted to Cartesian coordinates ***INLINE_MATH_53*** on an HTML5 canvas: ***DISPLAY_MATH_9 DISPLAY_MATH_10***

Each intensity cell is rendered using an active phosphor colormap:

- **Phosphor Amber Palette**: Dark bronze background (***INLINE_MATH_54***), transitioning to warm gold (***INLINE_MATH_55***) and brilliant yellow-white (***INLINE_MATH_56***).
- **Cyan Oceanic Palette**: Deep midnight navy (***INLINE_MATH_57***), transitioning to luminescent cyan (***INLINE_MATH_58***) and bright white (***INLINE_MATH_59***).

Concentric range circles (***INLINE_MATH_60***) and azimuth radial guides (***INLINE_MATH_61***) are overlaid with alpha blending, reproducing the exact operational appearance of commercial subsea topside consoles.

V. BENCHMARK VERIFICATION RESULTS

1. **Acoustic Range Resolution**: Range bin discretization tests verify that a ***INLINE_MATH_62*** sweep with ***INLINE_MATH_63*** bins resolves targets with ***INLINE_MATH_64*** precision, while a ***INLINE_MATH_65*** sweep resolves targets with ***INLINE_MATH_66*** precision.
2. **Transmission Loss Monotonicity**: Verifying two-way spherical spreading (***INLINE_MATH_67***) and chemical absorption confirms that echo returns from identical targets decay strictly with distance without numerical overflow or negative intensities.
3. **Shadow Occlusion Integrity**: Point-mass ray casting confirms that range cells immediately behind intersected objects register zero intensity (***INLINE_MATH_68***), producing true geometric shadows.
4. **Computational Performance**: Tracing 96 beams ***INLINE_MATH_69*** 128 range cells (***INLINE_MATH_70*** acoustic samples) and blitting to the canvas executes in ***INLINE_MATH_71*** per physics cycle on standard desktop hardware, sustaining full 60 FPS performance without dedicated WebGL compute shaders.

VI. CONCLUSION

Milestone 3.0 establishes real-time multibeam Forward-Looking Sonar simulation for web-native marine robotics. By deriving fan-beam geometry, physical two-way transmission loss, Lambertian acoustic scattering, geometric shadow casting, and polar-to-Cartesian scan conversion, the simulator provides subsea pilots and autonomous perception algorithms with physically grounded acoustic imagery in zero-visibility waters.

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